📝 Chemical Bonding — Assignment Questions

A. Multiple Choice Questions (MCQs)

1. Atoms form chemical bonds mainly to:
    1. Increase their mass
    2. Become electrically charged
    3. Achieve a stable electron configuration
    4. Increase their temperature
2. An ionic bond forms when atoms:
    1. Share electrons
    2. Transfer electrons
    3. Split electrons equally
    4. Form resonance structures
3. A covalent bond involves:
    1. Electron transfer
    2. A sea of delocalized electrons
    3. Sharing of electron pairs
    4. Attraction between ions
4. Metallic bonding explains why metals are:
    1. Brittle
    2. Poor conductors
    3. Malleable and shiny
    4. Non-reactive
5. A polar covalent bond occurs when:

1. Electrons are shared equally
2. Electrons are transferred
3. Electrons are shared unequally

4. Electrons are delocalized

B. Short Answer Questions

1. State the octet rule.
2. Why does bonding lower the energy of a system?
3. Define bond energy.
4. What is bond length?
5. What determines whether a bond is ionic or covalent?

C. Table Completion

1. Complete the table:

D. Conceptual Questions

2. Explain why Na and Cl form an ionic bond instead of a covalent bond.
3. Why are shorter bonds generally stronger?
4. Describe the difference between polar and nonpolar covalent bonds.
5. Give one example of an exception to the octet rule and explain why it is an exception.
6. What are resonance structures, and why are they needed?

E. Application / Problem-Solving

1. Predict the bond type (ionic, polar covalent, nonpolar covalent, metallic) for each pair:
2. H–H
3. H–O
4. Na–Cl
5. Fe–Fe
6. Explain why CO_2 is a covalent compound even though oxygen is more electronegative than carbon.
7. A molecule has very high bond energy. What does this tell you about its stability and bond length?

8. Metals conduct electricity even in solid form. Use metallic bonding to explain why.

F. Extended Response

1. Compare ionic, covalent, and metallic bonds in terms of:

- How they form
- Type of particles involved
- Strength
- Properties of the resulting substances

6. Describe how chemical bonding influences the structure and behavior of living organisms, materials, or everyday substances.